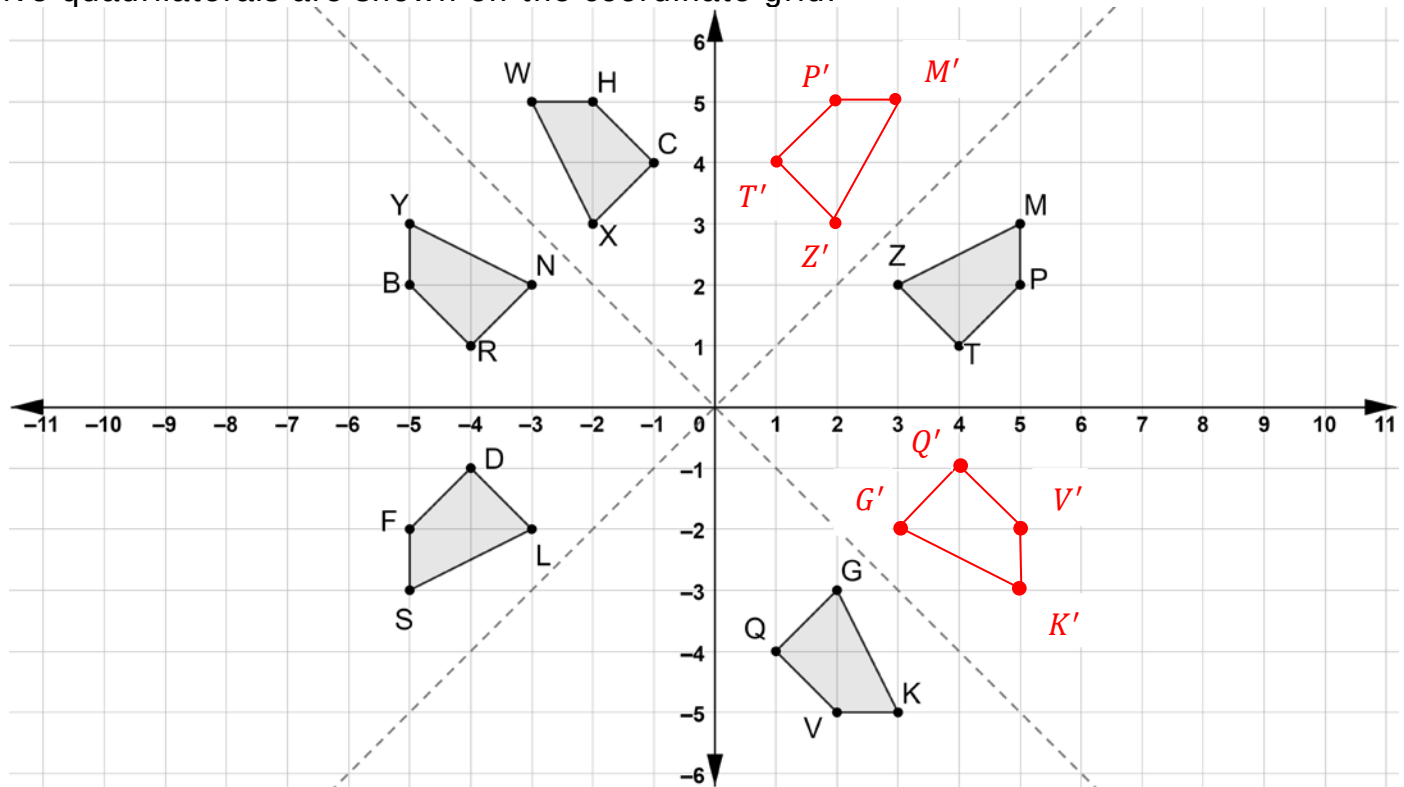


Five quadrilaterals are shown on the coordinate grid.



1. Draw the reflection of quadrilateral $MPTZ$ across the line $y = x$. Label the reflection as $M'P'T'Z'$.
2. Draw the reflection of quadrilateral $VKGQ$ across the line $y = -x$. Label the reflection as $V'K'G'Q'$.
3. Select all of the quadrilaterals that are reflections of quadrilateral $YBRN$.
 - ☒ Quadrilateral $WHCX$
 - ☒ Quadrilateral $MPTZ$
 - ☒ Quadrilateral $KVQG$
 - ☒ Quadrilateral $SFDL$
 - ☐ Quadrilateral $M'P'T'Z'$
 - ☐ Quadrilateral $K'V'Q'G'$
4. The algebraic description of the reflection of quadrilateral $BYNR$, the preimage, onto quadrilateral $PMZT$, the image, is $(x, y) \rightarrow (-x, y)$. Describe how the coordinates of the vertices of quadrilateral $BYNR$, the preimage, and the coordinates of quadrilateral $PMZT$, the image, can be used to determine the line of reflection.

$B: (-5, 2)$ $P: (5, 2)$

$Y: (-5, 3)$ $M: (5, 3)$

$N: (-3, 2)$ $Z: (3, 2)$

$R: (-4, 1)$ $T: (4, 1)$

The line of reflection is equidistant from the preimage coordinates to the image coordinates.

5. Which quadrilateral is the image of reflecting quadrilateral $DLSF$ across the x -axis?

$RNYB$

6. Which quadrilateral is the image of reflecting quadrilateral $M'P'T'Z'$ across the y -axis?

$WHCX$

7. Describe how the coordinates of the vertices of quadrilateral $BYNR$, the preimage, and the coordinates of quadrilateral $FSLD$, the image, can be used to determine the algebraic description of the reflection.

$B: (-5, 2)$	$F: (-5, -2)$	$(x, y) \rightarrow (x, -y)$
$Y: (-5, 3)$	$S: (-5, -3)$	
$N: (-3, 2)$	$L: (-3, -2)$	All x -coordinates are the same, the
$R: (-4, 1)$	$D: (-4, -1)$	y -coordinates are the opposite signs.

8. Describe how the coordinates of the vertices of quadrilateral $WHCX$, the preimage, and the coordinates of quadrilateral $YBRN$, the image, can be used to determine the algebraic description of the reflection.

$W: (-3, 5)$	$Y: (-5, 3)$	The coordinates change spots and signs.
$H: (-2, 5)$	$B: (-5, 2)$	$(x, y) \rightarrow (-y, -x)$
$C: (-1, 4)$	$R: (-4, 1)$	
$X: (-2, 3)$	$N: (-3, 2)$	

9. Write the algebraic description for the reflection of quadrilateral $BYNR$, the preimage, onto quadrilateral $PMZT$, the image.

$B: (-5, 2)$	$P: (5, 2)$	
$Y: (-5, 3)$	$M: (5, 3)$	x -coordinate changes signs.
$N: (-3, 2)$	$Z: (3, 2)$	$(x, y) \rightarrow (-x, y)$
$R: (-4, 1)$	$T: (4, 1)$	

10. Write the algebraic description for the reflection of quadrilateral $V'K'G'Q'$, the preimage, onto quadrilateral $HWXC$, the image.

$V': (5, -2)$	$H: (-2, 5)$	
$K': (5, -3)$	$W: (-3, 5)$	The x -coordinate and y -coordinate
$G': (3, -2)$	$X: (-2, 3)$	change spots.
$Q': (4, -1)$	$C: (-1, 4)$	$(x, y) \rightarrow (y, x)$